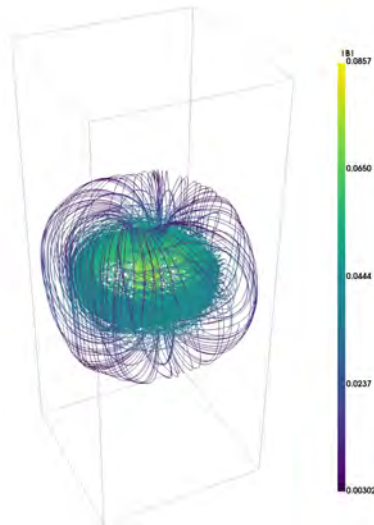


MFEM COMMUNITY WORKSHOP 2026

Structure-preserving, high-order, efficient MHD simulation with MFEM

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Outline

1. Fully divergence preserving, helicity, energy-conserving FEM for incompressible MHD

Mao & Xi (2025), *J. Comput. Phys.* 538, 114130.

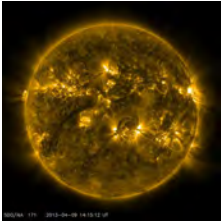
2. Structure-preserving, high-order ALE FEM for compressible ideal MHD systems

Mao & Xi (2026), *SIAM J. Sci. Comput.* 48, A1338–A1366.

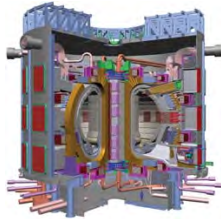
3. Structure-preserving, high-order Lagrangian scheme for magnetic relaxation in plasmas

Mao & Xi (2026), *SIAM J. Sci. Comput.*, accepted.

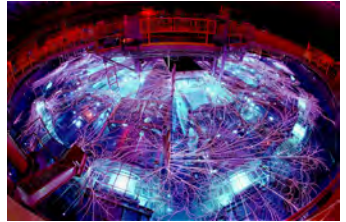
Magnetohydrodynamics (MHD)



Solar corona



Magnetic confinement fusion



Magnetically driven Z-pinch

MHD models have important applications in astrophysics, geophysics, and fusion.

Images: NASA/SDO; ITER rendering / Physics World; Sandia National Laboratories.

The MHD model

Hydrodynamics

Compressible

$$\begin{aligned}\partial_t \rho + \operatorname{div}(\rho \mathbf{u}) &= 0, \\ \partial_t(\rho \mathbf{u}) + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u}) + \nabla p &= \mathbf{J} \times \mathbf{B}, \\ \partial_t(\rho e) + \operatorname{div}(\rho e \mathbf{u}) + p \operatorname{div} \mathbf{u} &= 0.\end{aligned}$$

Incompressible

$$\begin{aligned}\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p - \frac{1}{Re} \Delta \mathbf{u} &= \mathbf{J} \times \mathbf{B}, \\ \operatorname{div} \mathbf{u} &= 0.\end{aligned}$$

Electromagnetics

$$\begin{aligned}\partial_t \mathbf{B} + \operatorname{curl} \mathbf{E} &= 0, \\ \mathbf{E} + \mathbf{u} \times \mathbf{B} &= Rm^{-1} \mathbf{J}, \\ \mathbf{J} &= \operatorname{curl}(\mathbf{B}/\mu_0), \\ \operatorname{div} \mathbf{B} &= 0.\end{aligned}$$

Ideal limit: $Rm \rightarrow \infty$.

Physical properties

Critical physical properties of the MHD model:

Solenoidal magnetic field $\operatorname{div} \mathbf{B} = 0$

Incompressibility $\operatorname{div} \mathbf{u} = 0$ (incompressible flow)

Current density conservation $\operatorname{div} \mathbf{J} = 0$

Energy law $\frac{d\mathcal{E}}{dt} = -Re^{-1}\|\nabla \mathbf{u}\|^2 - Rm^{-1}\|\mathbf{J}\|^2 \leq 0, \quad \mathcal{E} = \frac{1}{2}\|\mathbf{u}\|^2 + \frac{1}{2\mu_0}\|\mathbf{B}\|^2$

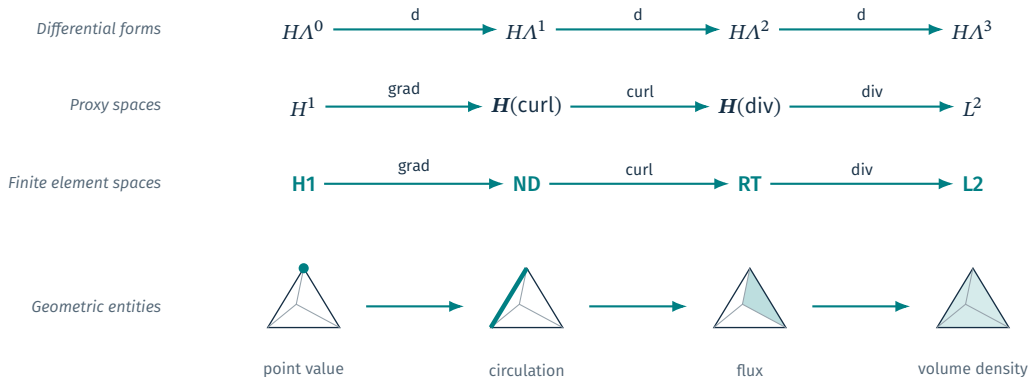
Magnetic helicity $\frac{d}{dt} \int_{\Omega} \mathbf{A} \cdot \mathbf{B} \, d\mathbf{x} = 0$ (ideal MHD), $\operatorname{curl} \mathbf{A} = \mathbf{B}$.

Fluid helicity conservation $\frac{d}{dt} \int_{\Omega} \mathbf{u} \cdot \boldsymbol{\omega} \, d\mathbf{x} = 0$ ($\mathbf{B} = 0$, inviscid), $\boldsymbol{\omega} = \operatorname{curl} \mathbf{u}$.

Density positivity $\rho > 0$

Preserving these physical properties is essential for reliable numerical simulations.

Finite Element Exterior Calculus (FEEC) in MFEM



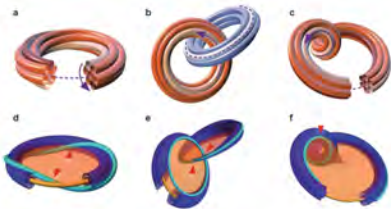
FEEC: Arnold et al. (2006), *Acta Numer.* 15, 1–155.

MFEM: Anderson et al. (2021), *Comput. Math. Appl.* 81, 42–74.

Magnetic helicity conservation

Magnetic helicity

$$\mathcal{H}_M = \int_{\Omega} \mathbf{A} \cdot \mathbf{B} \, d\mathbf{x}, \quad \text{curl } \mathbf{A} = \mathbf{B}.$$



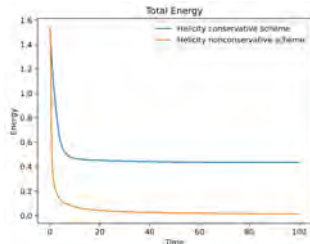
Helicity measures net linkage and twist of flux tubes; it is conserved in ideal MHD.

Helicity illustration: Scheeler et al. (2017), *Science* 357, 487–491.

Arnold inequality: Arnold & Khesin (2021), *Topological Methods in Hydrodynamics*, 2nd ed., Springer.

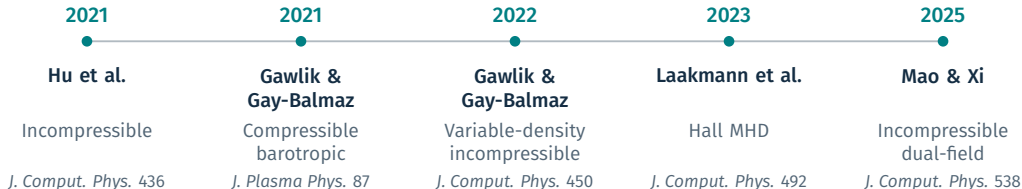
Arnold inequality

$$E_B = \frac{1}{2\mu_0} \|\mathbf{B}\|^2 \geq C_{\Omega} |\mathcal{H}_M|.$$



Nonzero conserved helicity prevents energy decay to an unphysical zero state.

Helicity-conserving methods



Previous methods are based on projections between FEEC spaces, requiring nonlinear solves at each time step.

$$\frac{d\mathcal{H}_M}{dt} = -2(\mathbf{E}, \mathbf{B}), \quad \mathbf{E} = -\mathbf{u} \times \mathbf{B} \implies (\mathbf{E}, \mathbf{B}) = 0.$$

At the discrete level:

$$\begin{aligned} \mathbf{H}_h &= \Pi_{\text{ND}} \mathbf{B}_h, & \mathbf{E}_h &= -\Pi_{\text{ND}}(\mathbf{u}_h \times \mathbf{H}_h), \\ (\mathbf{E}_h, \mathbf{B}_h) &= (\mathbf{E}_h, \mathbf{H}_h) = -(\mathbf{u}_h \times \mathbf{H}_h, \mathbf{H}_h) = 0 \implies \frac{d\mathcal{H}_{M,h}}{dt} = 0. \end{aligned}$$

Dual-field weak form

A dual-field weak formulation for incompressible MHD

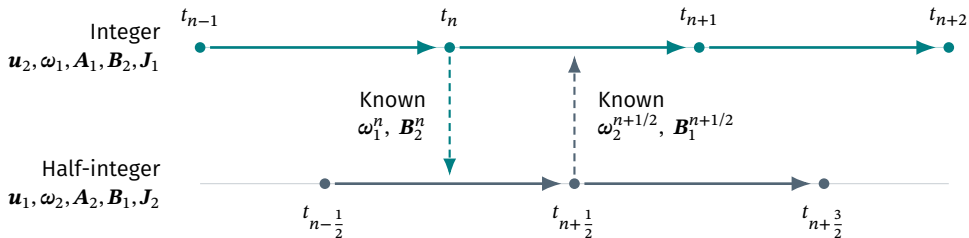
Scalar variables in H^1 and L^2 , vector variables in $H(\text{curl})$ and $H(\text{div})$.

$$\text{Formulation I} \quad \left\{ \begin{array}{l} (\dot{\mathbf{u}}_2, \mathbf{v}_2) + (\boldsymbol{\omega}_2 \times \mathbf{u}_2, \mathbf{v}_2) + Re^{-1}(\text{curl } \boldsymbol{\omega}_1, \mathbf{v}_2) - (P_3, \text{div } \mathbf{v}_2) - s(\mathbf{J}_1 \times \mathbf{B}_1, \mathbf{v}_2) = 0, \\ (\boldsymbol{\omega}_1, \mathbf{z}_1) = (\mathbf{u}_2, \text{curl } \mathbf{z}_1), \\ (\text{div } \mathbf{u}_2, q_3) = 0, \\ (\dot{\mathbf{A}}_1, \mathbf{d}_1) + (Rm^{-1} \mathbf{J}_1 - \mathbf{u}_2 \times \mathbf{B}_1, \mathbf{d}_1) = 0, \\ (\mathbf{B}_2, \mathbf{c}_2) = (\text{curl } \mathbf{A}_1, \mathbf{c}_2), \\ (\mathbf{J}_1, \mathbf{k}_1) = (\mathbf{B}_2, \text{curl } \mathbf{k}_1). \end{array} \right.$$

$$\text{Formulation II} \quad \left\{ \begin{array}{l} (\dot{\mathbf{u}}_1, \mathbf{v}_1) + (\boldsymbol{\omega}_1 \times \mathbf{u}_1, \mathbf{v}_1) + Re^{-1}(\boldsymbol{\omega}_2, \text{curl } \mathbf{v}_1) + (\nabla P_0, \mathbf{v}_1) - s(\mathbf{J}_2 \times \mathbf{B}_2, \mathbf{v}_1) = 0, \\ (\boldsymbol{\omega}_2, \mathbf{z}_2) = (\text{curl } \mathbf{u}_1, \mathbf{z}_2), \\ (\mathbf{u}_1, \nabla q_0) = 0, \\ (\dot{\mathbf{A}}_2, \mathbf{d}_2) + (Rm^{-1} \mathbf{J}_2 - \mathbf{u}_1 \times \mathbf{B}_2, \mathbf{d}_2) = 0, \\ (\mathbf{B}_1, \mathbf{c}_1) = (\mathbf{A}_2, \text{curl } \mathbf{c}_1), \\ (\mathbf{J}_2, \mathbf{k}_2) = (\text{curl } \mathbf{B}_1, \mathbf{k}_2). \end{array} \right.$$

Staggered Crank–Nicolson time stepping

With staggered Crank–Nicolson time stepping, each substep requires only a **linear solve**.



Discrete physical properties

Incompressibility

$$\operatorname{div} \mathbf{u}_{2,h}^n = 0$$

Solenoidal magnetic field

$$\operatorname{div} \mathbf{B}_{2,h}^n = 0$$

Current density conservation

$$\operatorname{div} \mathbf{J}_{2,h}^{n+1/2} = 0$$

Energy law

$$\delta_t E_{1,h} = -Re^{-1} \|\bar{\omega}_{2,h}\|^2 - sRm^{-1} \|\bar{\mathbf{J}}_{2,h}\|^2$$

$$\delta_t E_{2,h} = -Re^{-1} \|\bar{\omega}_{1,h}\|^2 - sRm^{-1} \|\bar{\mathbf{J}}_{1,h}\|^2$$

$$E_{1,h}, E_{2,h} \text{ conserved if } Re = Rm = \infty$$

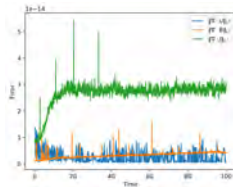
Magnetic helicity

$$\mathcal{H}_{M,h} \text{ conserved if } Rm = \infty$$

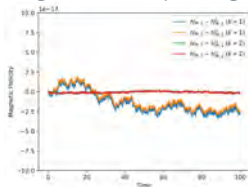
Fluid helicity

$$\mathcal{H}_{F,h} \text{ conserved if } \mathbf{B} = 0 \text{ and } Re = \infty$$

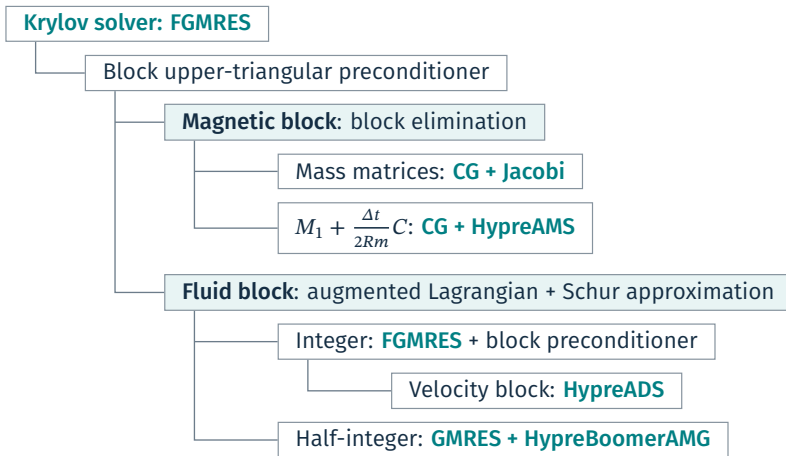
Divergence errors



Magnetic helicity change



Block preconditioning

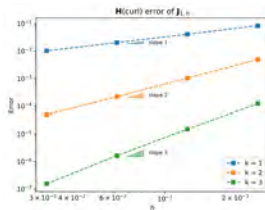
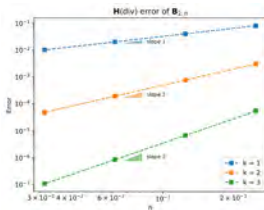
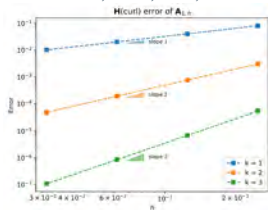


Augmented Lagrangian framework: Laakmann et al. (2022), *SIAM J. Sci. Comput.* 44, B1018–B1044.

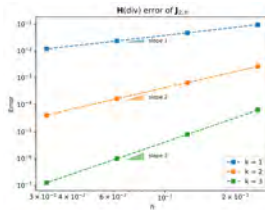
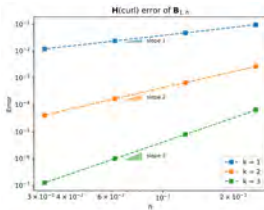
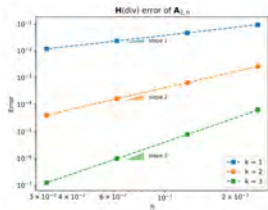
Spatial accuracy of the dual-field scheme

$k = 1, 2, 3$; $Re = Rm = s = 1$, $\Delta t = 0.01$, $T = 0.5$.

Integer sequence: $A_{1,h}$, $B_{2,h}$, $J_{1,h}$



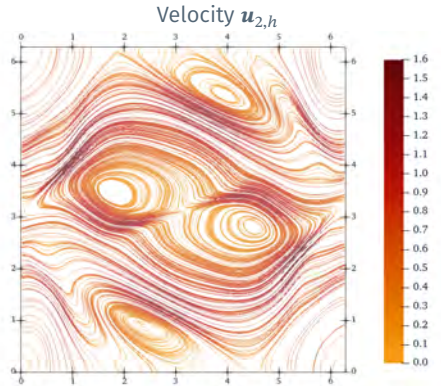
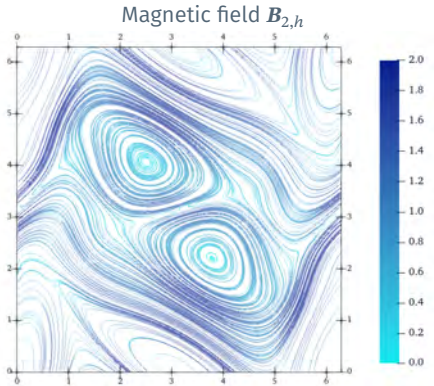
Half-integer sequence: $A_{2,h}$, $B_{1,h}$, $J_{2,h}$



Orszag–Tang vortex: magnetic and velocity fields

$Re = Rm = 100$, $200 \times 200 \times 25$ cells, $k = 1$, $t = 2$.

Periodic domain $(0, 2\pi)^2 \times (0, 1)$; section $z = 0.5$.

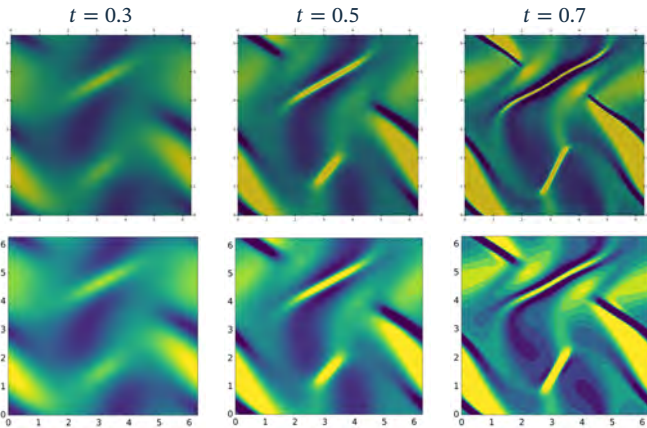


Orszag–Tang vortex: current-sheet evolution

$Re = Rm = 10^6$, $256 \times 256 \times 32$ cells, $k = 1$, $\Delta t = 0.025$.

Current density on $z = 0.5$.

Dual-field
FEEC

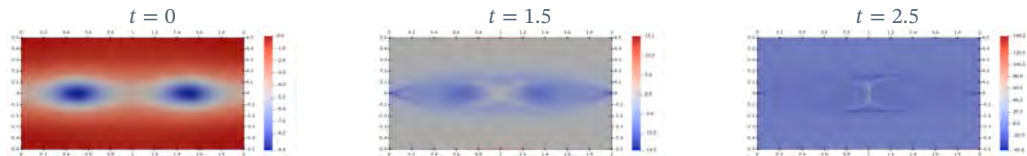


Bottom-row images: Kraus & Maj (2018), [arXiv:1707.03227v2](https://arxiv.org/abs/1707.03227v2).

Island coalescence

$Re = Rm = 10^5$, $s = 1$, $h = 1/160$, $k = 1$, $\Delta t = 0.005$.

Unknowns per system: 36,141,088 (integer); 36,141,448 (half-integer).



Average outer FGMRES iterations over all time steps:

Integer sequence				
$Re \backslash Rm$	10^2	10^3	10^4	10^5
10^2	7	7	8	9
10^3	7	9	10	10
10^4	8	9	11	11
10^5	8	10	12	13

Half-integer sequence				
$Re \backslash Rm$	10^2	10^3	10^4	10^5
10^2	29	33	34	34
10^3	37	44	46	46
10^4	48	60	60	61
10^5	51	62	63	66

Limitations

Some challenges remain:

Time order

These methods are at most second order in time. Helicity-preserving extension to higher order is nontrivial and expensive.

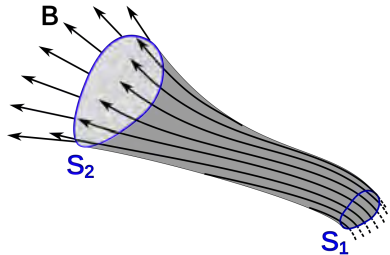
Topology preservation

Helicity $\mathcal{H}_M = \int_{\Omega} \mathbf{A} \cdot \mathbf{B} \, d\mathbf{x}$ is a global quantity integrated over the whole domain. Its conservation alone is not sufficient to preserve full field-line connectivity in local regions.

Alfvén's theorem: frozen-in magnetic flux

For ideal MHD, magnetic flux through every material surface is conserved:

$$\frac{d}{dt} \int_{S(t)} \mathbf{B} \cdot \mathbf{n} dS = 0, \quad S(t) = \varphi_t(S_0).$$



Differential-form transport $\beta = \mathbf{B} \cdot \mathbf{n} dS, \quad (\partial_t + \mathcal{L}_{\mathbf{u}})\beta = 0.$

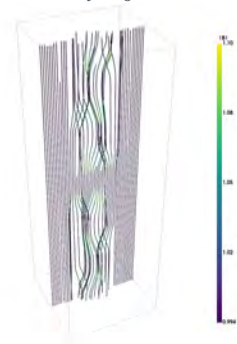
Piola map $JF^{-1}(\mathbf{B} \circ \varphi_t) = \mathbf{B}_0, \quad \mathbf{F} = \nabla_{\mathbf{x}} \varphi_t, J = \det \mathbf{F} > 0.$

Field lines move with the fluid, preserving their connectivity.

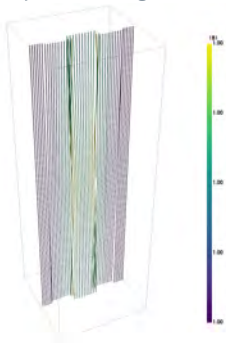
Alfvén (1943), *Ark. Mat. Astron. Fys.* 29, 1–7. Flux-tube diagram: Chetvorno, CC0, Wikimedia Commons.

Eulerian and Lagrangian methods

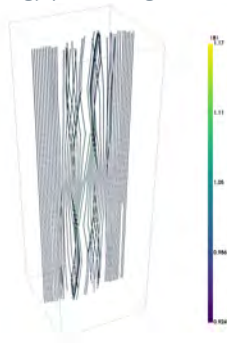
Initial
 $t = 0$



Eulerian: He et al. (2026)
Helicity conserving, $t = 2000$



Lagrangian: Mao & Xi (2026)
Topology preserving, $t = 2000$



Lagrangian methods naturally enforce $\mathbf{B} \circ \varphi_t = J^{-1} \mathbf{F} \mathbf{B}_0$, preserving full field-line topology.
Helicity conservation does not guarantee preservation of the full field-line topology.

He et al. (2026), *SIAM J. Sci. Comput.* 48, B165–B183.

Lagrangian and ALE methods for MHD

Reference	Method	Remarks
Rieben et al. (2007)	ALE	Resistive MHD; compatible FEM and divergence-preserving magnetic remap.
Robinson et al. (2011)	ALE	Ideal MHD; constrained transport and total-energy conservation.
Dobrev et al. (2012)	Lagrangian	High-order curvilinear hydrodynamics; Laghos miniapp .
Anderson et al. (2015)	ALE remap	Monotonic high-order DG remap. Remhos miniapp .
Anderson et al. (2018)	ALE	High-order multi-material hydrodynamics. Remhos miniapp
Nikl et al. (2022)	Lagrangian	High-order resistive MHD; Implemented in PETE2 with MFEM .
Boscheri et al. (2024)	Lagrangian	Ideal MHD; Piola-based divergence-preserving finite volumes.
Mao & Xi (2026)	ALE	High-order ideal MHD; helicity-conserving, divergence-preserving remap. Implemented in MFEM .

Basic idea in MHD: $\mathbf{B}$ as a discrete 2-form (e.g. RT space), Piola map: $\mathbf{B} \circ \varphi_t = J^{-1} \mathbf{F} \hat{\mathbf{B}}$.

Material derivatives for differential forms

$$\text{Material derivative: } D_{t,\mathbf{u}}^{(k)} = \partial_t + \mathcal{L}_{\mathbf{u}}$$

Form	Space	Quantity	Generalized material derivative	Piola map
0	H^1	Scalar a	$D_{t,\mathbf{u}}^{(0)} a = \partial_t a + \mathbf{u} \cdot \nabla a$	$a = \hat{a}$
1	$H(\text{curl})$	Circulation $\mathbf{A}$	$D_{t,\mathbf{u}}^{(1)} \mathbf{A} = \partial_t \mathbf{A} + (\text{curl } \mathbf{A}) \times \mathbf{u} + \nabla(\mathbf{u} \cdot \mathbf{A})$	$\mathbf{A} = \mathbf{J}_F^{-T} \hat{\mathbf{A}}$
2	$H(\text{div})$	Flux $\mathbf{B}$	$D_{t,\mathbf{u}}^{(2)} \mathbf{B} = \partial_t \mathbf{B} + \text{curl}(\mathbf{B} \times \mathbf{u}) + \mathbf{u} \text{ div } \mathbf{B}$	$\mathbf{B} = \frac{\mathbf{J}_F}{\det \mathbf{J}_F} \hat{\mathbf{B}}$
3	L^2	Density ρ	$D_{t,\mathbf{u}}^{(3)} \rho = \partial_t \rho + \text{div}(\rho \mathbf{u})$	$\rho = \frac{\hat{\rho}}{\det \mathbf{J}_F}$

The Lagrangian form of compressible MHD

Compressible ideal MHD

$$\partial_t \rho + \operatorname{div}(\rho \mathbf{u}) = 0,$$

$$\partial_t(\rho \mathbf{u}) + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u}) + \nabla p = \mu_0^{-1}(\operatorname{curl} \mathbf{B}) \times \mathbf{B},$$

$$\partial_t(\rho e) + \operatorname{div}(\rho e \mathbf{u}) + p \operatorname{div} \mathbf{u} = 0,$$

$$\partial_t \mathbf{B} + \operatorname{curl}(\mathbf{B} \times \mathbf{u}) = 0,$$

$$\operatorname{div} \mathbf{B} = 0, \quad p = (\gamma - 1)\rho e.$$

Geometric transport

$$D_{t,\mathbf{u}}^{(3)} \rho = 0, \quad D_{t,\mathbf{u}}^{(2)} \mathbf{B} = 0, \quad D_{t,\mathbf{u}}^{(1)} \mathbf{A} = 0.$$

(Lie gauge: $\phi = \mathbf{u} \cdot \mathbf{A}$, $\mathbf{B} = \operatorname{curl} \mathbf{A}$.)

Fluid dynamics

$$\rho D_{t,\mathbf{u}}^{(0)} \mathbf{u} = \operatorname{div}(\boldsymbol{\sigma} + \boldsymbol{\sigma}_B),$$

$$\rho D_{t,\mathbf{u}}^{(0)} e = \boldsymbol{\sigma} : \nabla \mathbf{u}.$$

$$\boldsymbol{\sigma} = -pI, \quad \boldsymbol{\sigma}_B = \mu_0^{-1}(\mathbf{B} \otimes \mathbf{B} - \frac{1}{2}|\mathbf{B}|^2 I).$$

Dobrev et al. (2012), *SIAM J. Sci. Comput.* 34, B606–B641. Nikl et al. (2022), *J. Comput. Phys.* 464, 111158.

Lie gauge: D'Avignon et al. (2016), *Phys. Plasmas* 23, 062101.

Finite element discretization

High-order FEEC discretization

Mesh position and velocity: continuous H1 elements.

Internal energy: discontinuous L2 elements with scalar (0-form) mapping.

$\mathbf{A}_h$: ND; $\mathbf{B}_h$: RT; ρ_h : L2 element with integral-preserving mapping.

Semi-discrete ODE system

$$\dot{\mathbf{x}} = \mathbf{u},$$

$$M_u \dot{\mathbf{u}} = -S\mathbf{1} - \mathbf{F}_B,$$

$$M_e \dot{\mathbf{e}} = S^T \mathbf{u}.$$

During a Lagrangian step, the coefficients of ρ_h , $\mathbf{A}_h$, and $\mathbf{B}_h$ remain fixed.

Hydrodynamic discretization: Dobrev et al. (2012), *SIAM J. Sci. Comput.* 34, B606–B641.

Helicity conservation

Vector fields: Piola transforms

$\mathbf{F} = \nabla_{\mathbf{X}} \mathbf{x}_h, J = \det \mathbf{F} > 0.$

$$\begin{aligned}\mathcal{H}_h(t) &= \int_{\Omega_t} \mathbf{A}_h \cdot \mathbf{B}_h \, d\mathbf{x} = \int_{\Omega_0} (\mathbf{F}^{-T} \mathbf{A}_h^0) \cdot (J^{-1} \mathbf{F} \mathbf{B}_h^0) J \, d\mathbf{X} \\ &= \int_{\Omega_0} \mathbf{A}_h^0 \cdot \mathbf{B}_h^0 \, d\mathbf{X} = \mathcal{H}_h(0).\end{aligned}$$

Differential forms: wedge product

$\alpha_h(t) \leftrightarrow \mathbf{A}_h(t), \beta_h(t) \leftrightarrow \mathbf{B}_h(t); \quad \mathbf{x}_h^* \alpha_h(t) = \alpha_h(0), \mathbf{x}_h^* \beta_h(t) = \beta_h(0).$

$$\begin{aligned}\mathcal{H}_h(t) &= \int_{\Omega_t} \alpha_h(t) \wedge \beta_h(t) = \int_{\Omega_0} \mathbf{x}_h^* (\alpha_h(t) \wedge \beta_h(t)) \\ &= \int_{\Omega_0} \mathbf{x}_h^* \alpha_h(t) \wedge \mathbf{x}_h^* \beta_h(t) = \int_{\Omega_0} \alpha_h(0) \wedge \beta_h(0) = \mathcal{H}_h(0).\end{aligned}$$

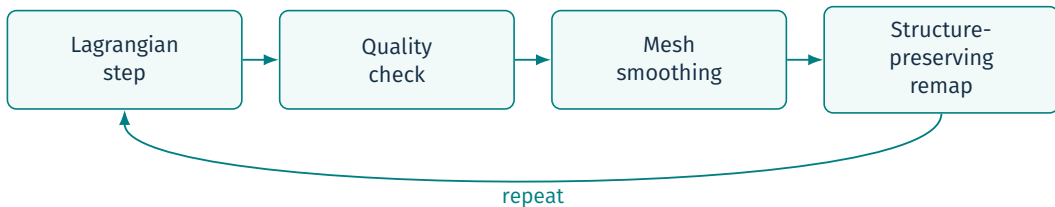
Helicity (and the whole topology) is naturally preserved at arbitrary time order during Lagrangian steps.

Structure-preserving ALE remapping

Mesh distortion limits the Lagrangian step

Moving with the fluid can stretch and twist elements, degrading accuracy and restricting the time step.

Arbitrary Lagrangian–Eulerian (ALE) method



Structure-preserving field transfer

Field	Remap	Preserved structure
ρ_h	Constrained L^2 projection with FCT	Mass and positive density
e_h	Upwind DG pseudo-advection	—
$\mathbf{u}_h$	DG transfer and Clément projection	—
$\mathbf{A}_h$	Auxiliary-variable midpoint update	Magnetic helicity
$\mathbf{B}_h$	Curl of the remapped potential	$\operatorname{div} \mathbf{B}_h = 0$

Helicity-preserving remap of $\mathbf{A}_h$

$$(\delta \mathbf{A}_h, \mathbf{C}_h) + (\mathbf{w}_h \times \mu_0 \mathbf{H}_h, \mathbf{C}_h) = 0, \quad (\mu_0 \mathbf{H}_h, \mathbf{K}_h) = (\operatorname{curl} \bar{\mathbf{A}}_h, \mathbf{K}_h).$$

$$\bar{\mathbf{A}}_h = \mathbf{A}_h^m + \frac{\Delta \tau}{2} \delta \mathbf{A}_h, \quad \mathbf{A}_h^{m+1} = \mathbf{A}_h^m + \Delta \tau \delta \mathbf{A}_h.$$

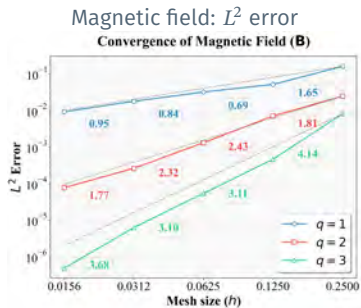
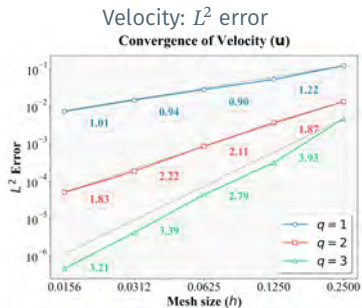
$$\mathbf{B}_h^{m+1} = \operatorname{curl} \mathbf{A}_h^{m+1}, \quad \operatorname{div} \mathbf{B}_h^{m+1} = 0, \quad \mathcal{H}_h^{m+1} = \mathcal{H}_h^m.$$

$$\delta \mathbf{A}_h, \mathbf{H}_h, \mathbf{C}_h, \mathbf{K}_h \in \text{ND}.$$

Hydrodynamic remap: Anderson et al. (2015), *Int. J. Numer. Meth. Fluids* 77, 249–273. Anderson et al. (2018), *SIAM J. Sci. Comput.* 40, B32–B58.

Taylor–Green vortex: high-order convergence

$q = 1, 2, 3$, $t = 0.5$, CFL = 0.5; remap every 10 Lagrangian steps.



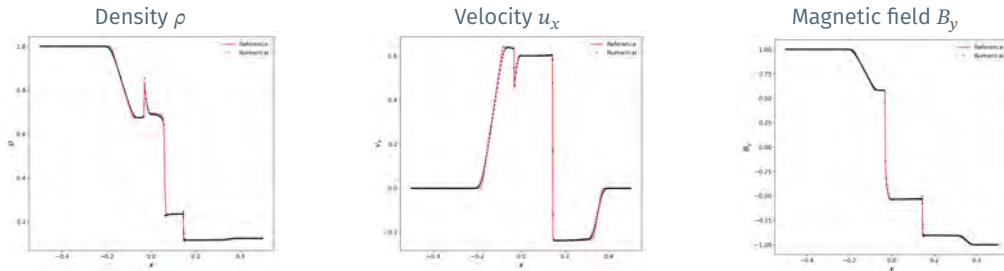
High-order spatial convergence is retained with repeated remapping.

Brio–Wu shock tube

400×10 quadrilaterals, $q = 2$, $t = 0.1$, $\gamma = 2$; Lagrangian solve with artificial viscosity.

Red curves: second-order MUSCL reference on 2,000 cells.

$$(\rho, p, B_y)_L = (1, 1, 1), \quad (\rho, p, B_y)_R = (0.125, 0.1, -1), \quad B_x = 0.75.$$



The profiles capture the MHD wave structure and the discontinuities well.

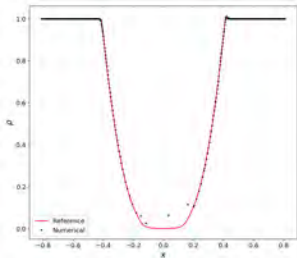
Benchmark: Brio & Wu (1988), *J. Comput. Phys.* 75, 400–422.

Near-vacuum expansion

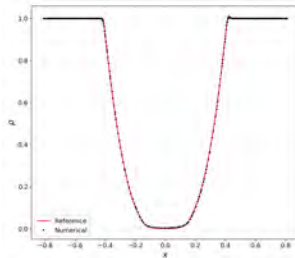
$400 \times 10 \times 10$ hexahedra, $q = 2$, $t = 0.1$; $\gamma = 5/3$; artificial viscosity.

$$(\rho, u_x, p, B_y)_{L/R} = (1, \mp 3.1, 0.45, 0.5).$$

Pure Lagrangian: density



ALE: density



ALE resolves the central rarefaction more accurately while keeping density positive.

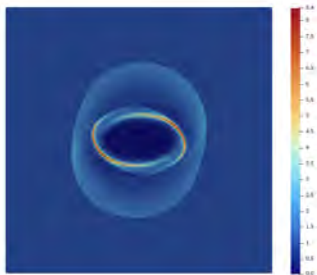
Benchmark: Fuchs et al. (2009), *J. Comput. Phys.* 228, 641–660.

MHD rotor: density and magnetic field

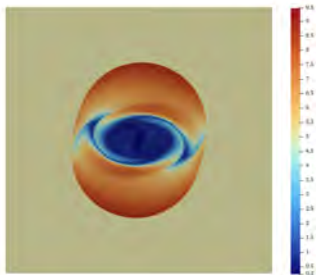
Dense rotating core in a lighter fluid; 320^2 quadrilaterals, $q = 2$, $t = 0.15$.

$\Omega = (-1, 1)^2$, $\rho_{\text{core}}/\rho_{\text{ambient}} = 10$, $p = 1$, $\gamma = 1.4$.

ALE density ρ_h



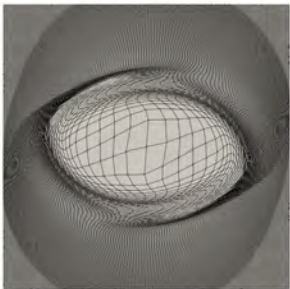
ALE magnetic magnitude $|\mathbf{B}_h|$



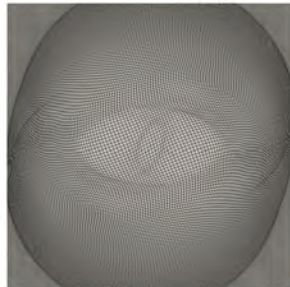
Mesh control in the MHD rotor

320^2 quadrilaterals, $q = 2$, $t = 0.15$.

Pure Lagrangian mesh

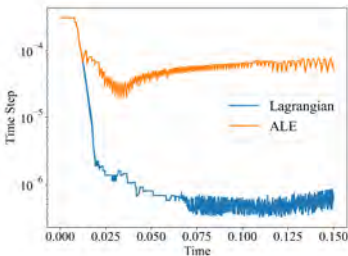


ALE mesh



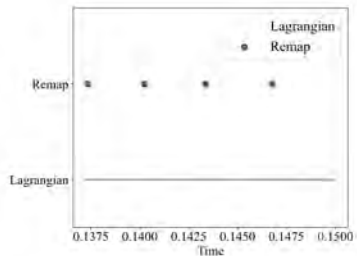
MHD rotor: time steps and adaptive remapping

320^2 quadrilaterals, $q = 2$, $t \leq 0.15$.



Larger time steps

ALE improves the time-step size by approximately one order of magnitude.



Infrequent remapping

Only four remaps in the final 200 Lagrangian steps, about one per 50 steps.

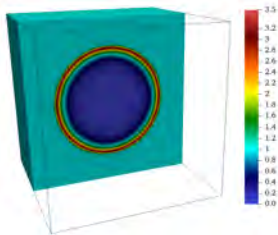
3D MHD blast

Point-energy release at the origin; 48^3 hexahedra, $q = 2$, $t = 0.25$.

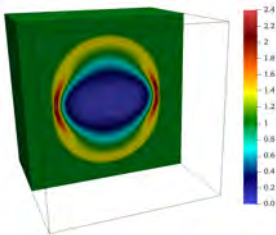
$\Omega = (-1, 1)^3$; section $z = 0$; CFL = 0.5.

$$\rho_0 = 1, \quad \mathbf{u}_0 = 0, \quad \gamma = 5/3, \quad \mathbf{B}_0 = (\beta\sqrt{4\mu_0/3}, 0, 0), \quad \mu_0 = 4\pi.$$

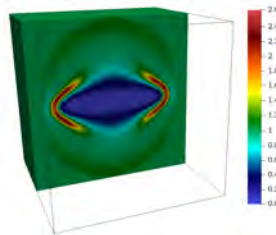
Density, $\beta = 0$



Density, $\beta = 1$



Density, $\beta = 2$

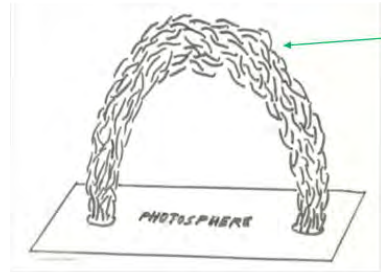


A stronger background magnetic field makes the blast increasingly anisotropic.

Magnetic relaxation and the Parker problem

Magnetic relaxation is a fundamental plasma process related to coronal heating.

- In the low- β , highly conducting corona, photospheric motions braid magnetic fields and store energy.
- Ideal magnetic relaxation lowers magnetic energy while preserving field-line topology, seeking a smooth force-free equilibrium.
- **Parker's conjecture:** generic ideal relaxation forms tangential discontinuities (current sheets).



Parker's sketch, reproduced in Low (2023), *Phys. Plasmas* 30, 012903.

Parker (1972), *Astrophys. J.* 174, 499.

Pontin & Hornig (2020), *Living Rev. Sol. Phys.* 17, 5.

The ideal magnetic-relaxation model

$$\begin{aligned}\partial_t \mathbf{B} + \operatorname{curl} \mathbf{E} &= 0, & \mathbf{E} &= -\mathbf{u} \times \mathbf{B}, \\ \mathbf{u} &= \tau \mathcal{A}(\mathbf{j} \times \mathbf{B}), & \mathbf{j} &= \operatorname{curl} \mathbf{B}, & \operatorname{div} \mathbf{B} &= 0.\end{aligned}$$

$\tau > 0$; $\mathcal{A}$ is positive semidefinite.

- $\mathcal{A} = I$: L^2 **gradient flow** (magneto-frictional).
- $\mathcal{A} = (I - \Delta)^{-1}$: H^1 **gradient flow**.

Topology preservation

$$\mathbf{B} \circ \varphi_t = J^{-1} F \mathbf{B}_0.$$

Field-line connectivity is transported by the flow map.

Magnetic-energy dissipation

$$\dot{\mathcal{E}} = -\tau(\mathbf{j} \times \mathbf{B}, \mathcal{A}(\mathbf{j} \times \mathbf{B})) \leq 0.$$

$\mathcal{E} = \frac{1}{2} \|\mathbf{B}\|^2$ decreases during relaxation.

Existing magnetic relaxation schemes

Method	$\text{div } \mathbf{B} = 0$	Energy dissipation	Full field-line topology
Craig & Sneyd (1986), Lagrangian FD	✓	✗	✓
Pontin et al. (2009), Stokes-based Lagrangian	✓	✗	✓
Candelaresi et al. (2014), mimetic Lagrangian	✓	✗	✓
He et al. (2026), Eulerian FE	✓	✓	✗
Farrell et al. (2026), Eulerian FE	✓	✓	✗
Mao & Xi (2026), Lagrangian FEEC + AVF	✓	✓	✓

Lagrangian relaxation

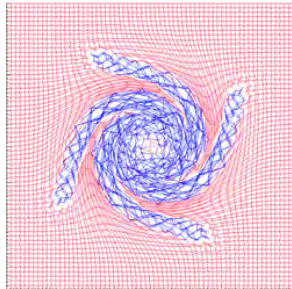
$$\mathbf{u}_h = \mathbf{j}_h \times \mathbf{B}_h = (\text{curl}_h \mathbf{B}_h) \times \mathbf{B}_h$$

$$\mathbf{B}_h \circ \mathbf{x}_h = J_h^{-1} F_h \mathbf{B}_h^0$$

Lagrangian relaxation schemes can exactly preserve topology, but can lose robustness.

- Severe **mesh distortion** and concave cells can cause accuracy loss or breakdown.
- **Energy dissipation** cannot be guaranteed.
- Extension to **high-order meshes** is nontrivial.

Mimetic method (GLEMUR): Candelaresi et al. (2014), *SIAM J. Sci. Comput.* 36, B952–B968.



GLEMUR: mimetic Pontin09 run; breakdown at $t \approx 238.2$.

From magnetic energy to dissipative relaxation

First variation of energy $\longleftrightarrow$ Weak Lorentz force

Magnetic energy on the reference mesh

$$\mathcal{E}_h(\mathbf{x}_h) = \frac{1}{2} \int_{\Omega_0} J_h^{-1} |F_h \mathbf{B}_h^0|^2 d\mathbf{X}.$$

First variation of magnetic energy

$$D\mathcal{E}_h(\mathbf{x}_h)[\mathbf{v}_h] = \int_{\Omega_t} (\mathbf{B}_h \otimes \mathbf{B}_h - \frac{1}{2} |\mathbf{B}_h|^2 I) : \nabla \mathbf{v}_h d\mathbf{x}.$$

Magnetic relaxation is then a gradient flow

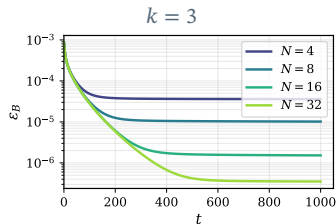
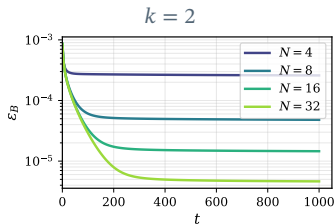
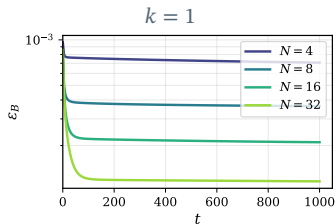
$$(\dot{\mathbf{x}}_h, \mathbf{v}_h)_H = -\tau D\mathcal{E}_h(\mathbf{x}_h)[\mathbf{v}_h].$$

$$\mathbf{v}_h = \dot{\mathbf{x}}_h \implies \frac{d\mathcal{E}_h}{dt} = D\mathcal{E}_h[\dot{\mathbf{x}}_h] = -\tau^{-1} \|\dot{\mathbf{x}}_h\|_H^2 \leq 0.$$

- $H = L^2$: $\mathcal{A} = I \longrightarrow L^2$ gradient flow.
- $H = H^1$: $\mathcal{A} = (I - \Delta)^{-1} \longrightarrow H^1$ gradient flow.

IsoHelix: convergence to the uniform field

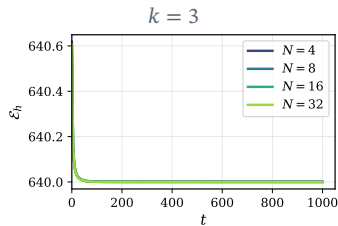
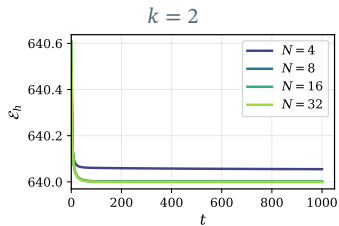
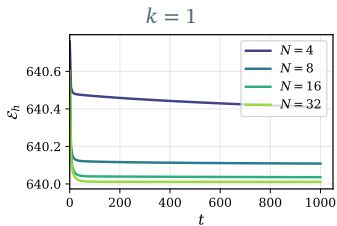
$N = 4, 8, 16, 32$; $N \times N \times 2.5N$ cells; equilibrium $\mathbf{B} = B_0 \mathbf{e}_z$



Higher-order elements reduce the error in the relaxed magnetic field.

IsoHelix: magnetic-energy decay

$N = 4, 8, 16, 32$; $N \times N \times 2.5N$ cells; same runs as the field-error curves.

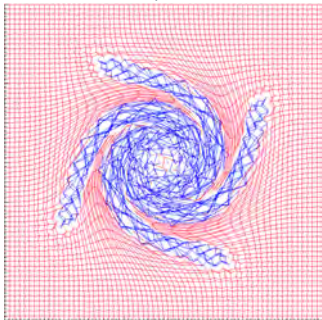


Magnetic energy decreases toward the uniform-field equilibrium.

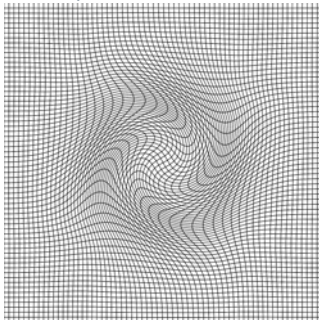
Pontin09: comparison under mesh distortion

$N = 64$; twists $\pm\pi$; section $z = 0$.

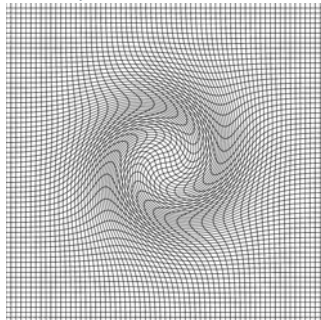
Mimetic, $t \approx 238.2$



Proposed $k = 1$, $t = 1000$



Proposed $k = 2$, $t = 1000$

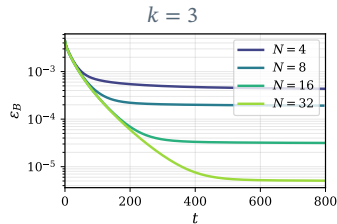
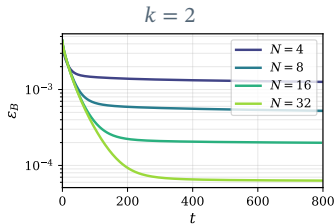
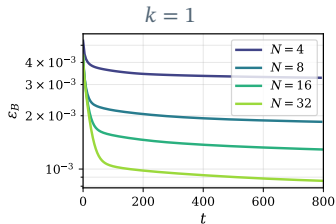


The mimetic run breaks down; the proposed method reaches the final time.

Mimetic method (GLEMUR): Candelaresi et al. (2014), *SIAM J. Sci. Comput.* 36, B952–B968.

Pontin09: robustness across all settings

Pontin09: twists $\pm\pi$; $N = 4, 8, 16, 32$.



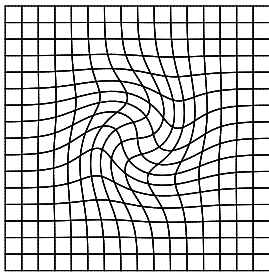
The twisted Pontin09 field relaxes toward the known uniform state.

Benchmark: Pontin et al. (2009), *Astrophys. J.* 700, 1449.

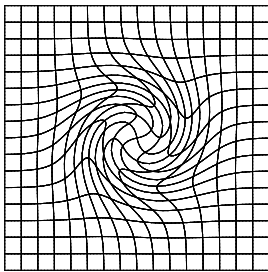
Pontin09: high-order meshes at stronger twists

$16 \times 16 \times 40$ cells, $k = 3$, $t = 1000$; section $z = 0$.

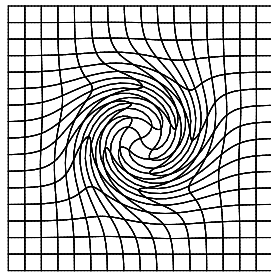
$$\phi_{1,2} = \pm\pi/2$$



$$\phi_{1,2} = \pm\pi$$



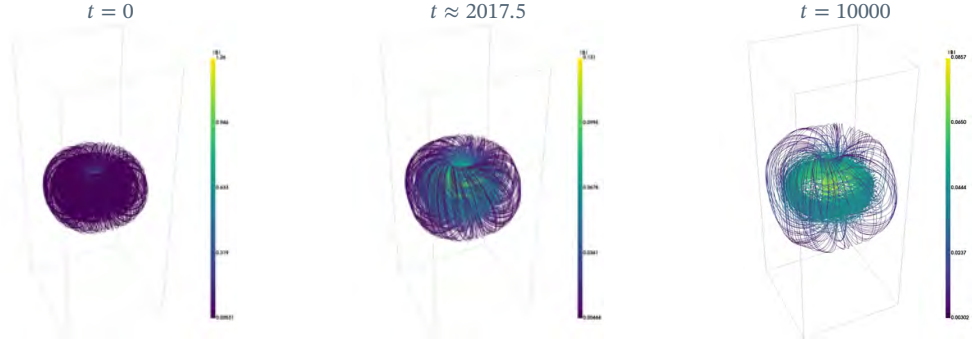
$$\phi_{1,2} = \pm 3\pi/2$$



Curvilinear elements accommodate severe distortion in these tests.

Hopf fibration: field-line evolution

Winding numbers $(3, 2)$, $s = 1$; streamlines crossing $z = 0$.

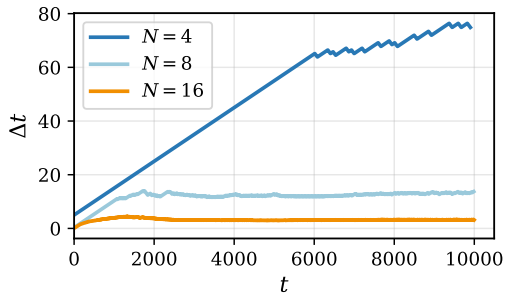


The field expands and weakens while retaining its linked structure.

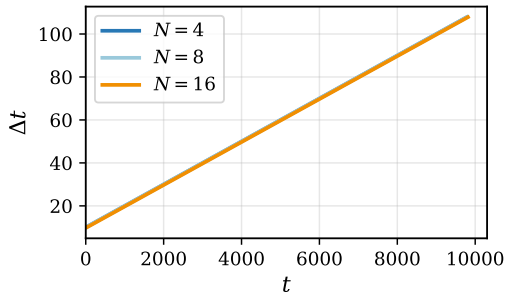
The relaxation metric and efficiency

Hopf fibrations; adaptive steps depend on Anderson–Picard convergence.

L^2 gradient flow



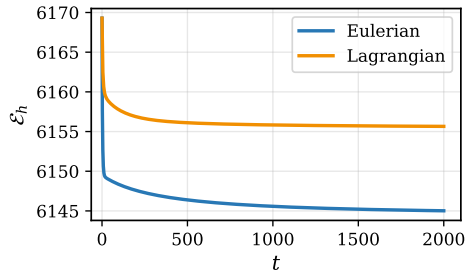
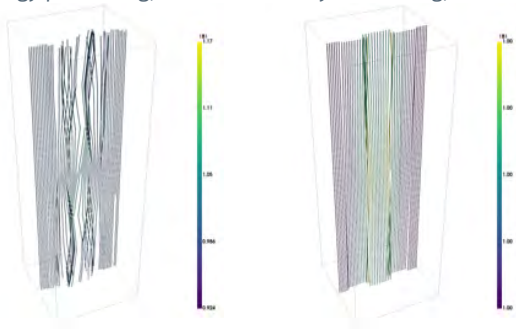
H^1 gradient flow



The H^1 flow is more robust in Picard convergence.

A braided field retains nontrivial structure

Lagrangian: Mao & Xi (2026) Eulerian: He et al. (2026)
Topology preserving, $t = 2000$ Helicity conserving, $t = 2000$



Helicity conservation alone permits a different relaxed state

He et al. (2026), *SIAM J. Sci. Comput.* 48, B165–B183.

Conclusion

1. Eulerian dual-field method for incompressible MHD

- The first linear helicity-preserving Eulerian scheme with a **dual-field formulation**.
- Preserves **divergence constraints**, the **energy law**, and **magnetic and fluid helicities**.
- Block preconditioners remain **robust at high Re and Rm up to 10^5 – 10^6** .

2. ALE method for compressible ideal MHD

- A clean Lagrangian formulation of MHD based on **material derivatives of differential forms**.
- FEEC transport naturally preserves **helicity** at **arbitrary temporal order** in Lagrangian steps.
- **Structure-preserving remaps** preserve helicity, $\operatorname{div} \mathbf{B}_h = 0$ and positive density.

3. Lagrangian magnetic relaxation

- The first relaxation scheme to preserve **topology** and guarantee **energy dissipation**.
- Reformulation as L^2 and H^1 **gradient flows** based on the first variation of the discrete energy.
- Robust and efficient relaxation with **high-order elements** and **matrix-free GPU** computation.

Thank you!

Related publications:

Shipeng Mao and Ruijie Xi (2025)

An incompressibility, $\operatorname{div} B = 0$ preserving, current density, helicity, energy-conserving finite element method for incompressible MHD systems

Journal of Computational Physics 538, 114130.

Shipeng Mao and Ruijie Xi (2026)

A structure-preserving, helicity-conserving high-order ALE finite element method for compressible ideal MHD systems

SIAM Journal on Scientific Computing 48(3), A1338–A1366.

Shipeng Mao and Ruijie Xi (2026)

A topology preserving, energy dissipative, high-order Lagrangian finite element method for magnetic relaxation in plasmas

SIAM Journal on Scientific Computing, accepted.